

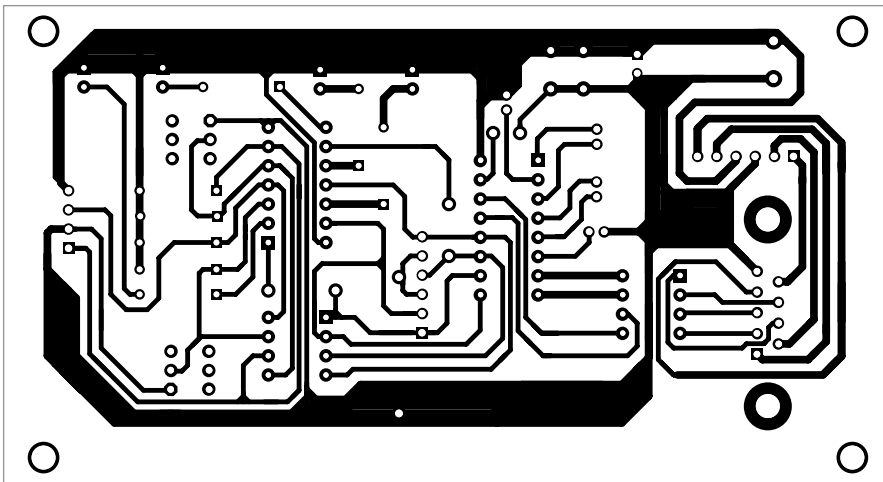


Fig. 2: Actual-size PCB pattern of configurable RS232 to TTL to I2C adaptor

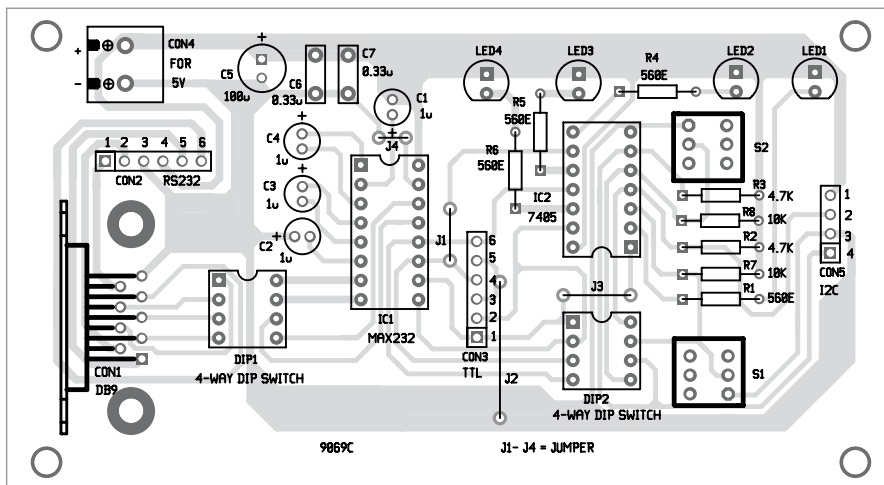


Fig. 3: Component layout of the PCB

tions for TTL to I2C converter built around inverters N2 and N3 of IC2. I2C signals are available on CON5.

Vs+ and Vs- at pins 2 and 6 respectively of IC1 are the two test points where the voltage levels obtained should be around 9V. These voltages are produced internally by MAX232 and should be within speci-

fied limits as stated in the data sheet.

The circuit requires a 5V DC regulated power supply and below 100mA of current to operate. You may use a small transformer providing around 10V/100mA and a 7805 regulator for the power supply. Regulator IC should be mounted on a heat-sink having thermal resis-

tance below 30°C/W.

You can provide two appropriate connectors at the power supply output: one for the adaptor circuit and another for the digital or embedded circuit for interfacing TTL or I2C signals.

Construction and testing

An actual-size, single-side PCB pattern of configurable RS232 to TTL to I2C adaptor circuit is shown in Fig. 2 and its component layout in Fig. 3. After mounting all components on the PCB, check proper connections again in the circuit. Connect the 5V DC power supply source to CON4 to power on the circuit.

Next, connect connector DB9 to the computer and send digital signals from serial communication software (for example, HyperTerminal) to the circuit. If none of the LEDs glow, check proper connections on DIP1 and DIP2 and ensure that transmit (pin 3) and receive (pin 2) pins of DB9 are connected to the appropriate pins. Otherwise, toggle S1 and S2; LED1 and LED2 should glow, respectively.

If everything is working fine, you may transmit any digital signals, either TTL or I2C or RS232, from DB9 to CON3 and CON5. **EFY**



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